



ONLINE TEST SERIES CONTACT PROGRAMME
(Academic Session : 2024 - 2025)

JEE (Main)
MINOR
10-11-2024

JEE(Main) : ENTHUSE + LEADER COURSE (PHASE : 3)
ANSWER KEY

PART 1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	B	A	D	D	C	B	A	A	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	C	B	D	A	C	A	C	D	B
SECTION-II	Q.	1	2	3	4	5					
	A.	3	5	412	1	582					

PART 2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	A	B	C	D	C	D	C	D	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	B	C	D	A	B	C	B	C	A
SECTION-II	Q.	1	2	3	4	5					
	A.	1	2	6	4	2					

PART 3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A	C	B	B	D	C	B	D	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	C	D	C	B	B	A	B	D	A
SECTION-II	Q.	1	2	3	4	5					
	A.	1	2	20	3	6					

HINT – SHEET

PART 1 : PHYSICS

SECTION-I

1. **Ans (A)**

Ist case $\rightarrow n = 60^\circ$ of $v_{\max}$

$$v = \frac{60 \times A\omega}{100}$$

$$n = 0.6 A\omega$$

$$\omega\sqrt{A^2 + x_1^2} = 0.6A\omega$$

$$A^2 + x_1^2 = 0.36A^2$$

$$x_1^2 = 0.64A^2$$

$$x_1 = 0.8A$$

IInd case $\rightarrow n = 80\%$ of $v_{\max}$

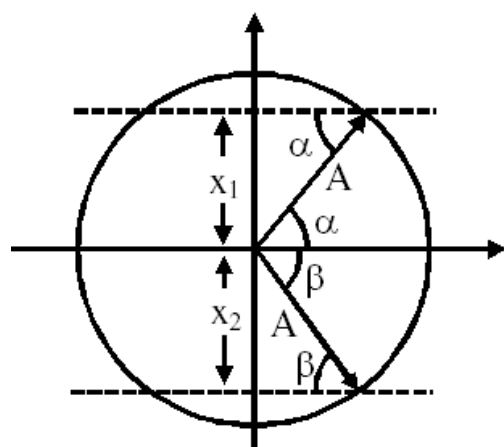
$$v = \frac{80 \times A\omega}{100} = 0.8A\omega$$

$$\omega\sqrt{A^2 - x_2^2} = 0.8A\omega$$

$$\omega^2(A^2 - x_2^2) = 0.64A^2\omega^2$$

$$x_2^2 = 0.36A^2$$

$$x_2 = 0.6A$$



$$\sin \alpha = \frac{x_1}{A} = \frac{0.8A}{A}$$

$$\sin \alpha = 0.8 = \frac{4}{5} = \sin 53^\circ$$

$$\alpha = 53^\circ$$

$$\sin \beta = \frac{x_2}{A} = \frac{0.6A}{A}$$

$$\sin \beta = 0.6 = \frac{3}{5} = \sin 37^\circ$$

$$\beta = 37^\circ$$

$$\theta = \omega t$$

$$(\alpha + \beta) = \omega t \Rightarrow t = \frac{90^\circ}{\omega} = \frac{\pi \times T}{2 \times 2\pi} = \frac{T}{4}$$

2. **Ans (B)**

$$(x - A) = B\sin \omega t$$

SHM of particle about $x = A$ and its amplitude is B .

3. **Ans (A)**

$$T = 2\pi\sqrt{\frac{M_{\text{total}}}{k_{\text{eff}}}}$$

$$k_{\text{eff}} = k_1 + k_2 \quad \text{Parallel combination}$$

$$= 2k + k = 3k$$

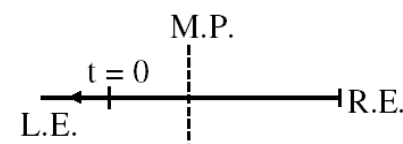
$$T = 2\pi\sqrt{\frac{3m}{3k}}$$

$$T = 2\pi\sqrt{\frac{m}{k}}$$

4. **Ans (D)**

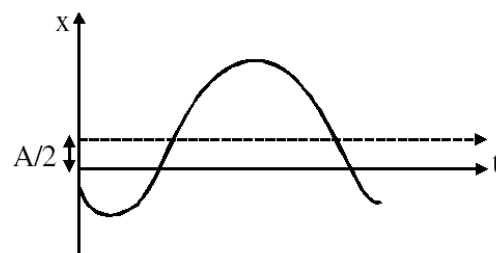
$$\text{Mean position at } x = \frac{A}{2}$$

$$\text{at } t = 0 \quad x = -\frac{3A}{4}$$



at $t = 0$ it will move towards left extreme

(negative displacement)



5. **Ans (D)**

$a = -\beta (x - 2)$ Compares with the standard equation of SHM $a = -\omega^2 x$.

$$\therefore \omega = \sqrt{\beta} \quad (\text{Equilibrium position } x = 2)$$

$$\therefore T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{\beta}}$$

6. **Ans (C)**

The maximum K.E. of ejected photoelectron

$$\text{is } (K.E.)_{\max} = h\nu - \phi_0$$

If the frequency of photon is doubled,

maximum kinetic energy of photon electron

$$\text{becomes } (K.E.)'_{\max} = 2h\nu - \phi_0$$

$$\frac{(K.E.)'_{\max}}{(K.E.)_{\max}} = \frac{2(h\nu - \frac{\phi_0}{2})}{h\nu - \phi_0} > 2$$

$$\text{Photo current} \propto \frac{\text{intensity of beam}}{h\nu}$$

If intensity and frequency both are doubled, the photocurrent remains same.

7. **Ans (B)**

$$\Delta Q = BE_p - BE_R$$

$$= 4 \times 7.2 - 4 \times 1.125$$

$$= 28.8 - 4.5 = 24.3 \text{ MeV}$$

8. **Ans (A)**

$$\frac{hc}{\lambda} - \phi = \frac{1}{2}mv^2 = \frac{p^2}{2m}$$

$$p = \frac{h}{\lambda_d}$$

$$\frac{hc}{\lambda} - \phi = \frac{h^2}{2m\lambda_d^2}$$

$$-hc \frac{\Delta\lambda}{\lambda^2} = \frac{h^2}{2m} \frac{(-2)}{\lambda_d^3} \Delta\lambda_d$$

$$-\frac{hc \cdot 2m}{h^2(-2)} = \frac{\lambda_d^3}{\lambda^2} = \frac{\Delta\lambda_d}{\Delta\lambda}$$

9. **Ans (A)**

$$E = 13.6 \times \frac{9}{4} = 30.6 \text{ eV}$$

10. **Ans (B)**

$$E_C - E_B = \frac{hc}{\lambda_1} \quad \dots(1)$$

$$E_B - E_A = \frac{hc}{\lambda_2} \quad \dots(2)$$

$$E_C - E_A = \frac{hc}{\lambda_3} \quad \dots(3)$$

On add equation (1) and (2)

$$E_C - E_A = hc \left(\frac{1}{\lambda_1} + \frac{1}{\lambda_2} \right)$$

$$\frac{hc}{\lambda_3} = hc \left(\frac{1}{\lambda_1} + \frac{1}{\lambda_2} \right) \Rightarrow \lambda_3 = \frac{\lambda_1 \lambda_2}{\lambda_1 + \lambda_2}$$

11. **Ans (B)**

Let n be number of fissions per second.

Each fission produces 200 MeV .

$n \times 200 \times 10^6 \text{ eV}$ is produced in one second by n fissions

$$\text{But } 1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$$

Hence, power produce

$$n \times 200 \times 10^6 \times 1.6 \times 10^{-19} \text{ Joule per second.}$$

Also $1 \text{ J/s} = 1 \text{ W}$ and

$$1000 \text{ J/s} = 1 \text{ kW.}$$

The required power is 1 kW .

$$\text{Hence, } \frac{n \times 200 \times 10^6 \times 1.6 \times 10^{-19}}{10^3} = 1$$

$$n = \frac{10^3}{2 \times 1.6 \times 10^{-11}} = \frac{10^{14}}{3.2}$$

$$= \frac{10}{3.2} \times 10^{13} = 3.125 \times 10^{13}$$

12. **Ans (C)**

The range of β particle lie between zero to some maximum value. During beta decay,

electrons or photons are released. Because a neutron or an antineutron is emitted simultaneously, there is an emission spectrum of electrons or positrons depending on the ratio or Q reaction energies boron by the large particle. The shape of this energy curve can be predicted from the Fermi theory of beta decay.

13. **Ans (B)**

Spectral lines of Balmer series are in visible range.

Balmer series lines are obtained due to transition of an electron to $n = 2$ state from any higher state.

14. **Ans (D)**

$$\lambda = \frac{h}{p}$$

$$\frac{\lambda_p}{\lambda_e} = \frac{p_e}{p_p} = \frac{m_e v_e}{m_p v_p}$$

$$2 = \frac{m_e}{m_p} \left(\frac{v_e}{4v_e} \right)$$

$$\therefore m_p = \frac{m_e}{8}$$

15. Ans (A)

$$\frac{1}{\lambda} = RZ^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \quad z = 2$$

$$\frac{1}{\lambda} = 4R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

$$\frac{1}{\lambda} = R \left(\frac{1}{\frac{n_1^2}{4}} - \frac{1}{\frac{n_2^2}{4}} \right)$$

$$\frac{n_1^2}{4} \text{ \& } \frac{n_2^2}{4} \text{ must be integer \& square terms}$$

$$n = 4 \text{ to } n = 2$$

16. Ans (C)

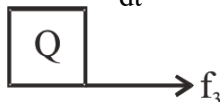
Total energy of SHM (After impulse)

$$m\omega^2 A^2 = \frac{1}{2} m\omega^2 A_1^2$$

$$\therefore A_1 = \sqrt{2}A$$

17. Ans (A)

$$-kx = \frac{3md^2x}{dt^2} \quad \dots (i)$$



$$f_3 = \frac{2md^2x}{dt^2} \quad \dots (ii)$$

18. Ans (C)

$$T = 2\pi \sqrt{\frac{l}{Mgd}} = 2\pi \sqrt{\frac{2ml\alpha}{2mg \cos \frac{\alpha}{2}}}$$

$$= 2\pi \sqrt{\frac{l}{g \cos \left(\frac{\alpha}{2} \right)}}$$

19. Ans (D)

$$\lambda = \frac{h}{\sqrt{2mE}}$$

$$\lambda' = \frac{h}{\sqrt{2m(E-V)}}$$

20. Ans (B)

$$\text{Kinetic energy of ball} = \frac{h^2}{2m\lambda^2}$$

$$\text{Loss in kinetic energy} = \frac{h^2}{2m} \left(\frac{1}{\lambda_1^2} - \frac{1}{\lambda_2^2} \right)$$

PART 1 : PHYSICS

SECTION-II

1. Ans (3)

$$\lambda = \frac{h}{mv}$$

$$\text{velocity of an } e^- \text{ in } \propto \frac{Z}{h}$$

H- like atom

$$\lambda \propto \frac{1}{v} \text{ so } \lambda \propto \frac{k}{v} \quad (k = \text{constant})$$

$$\frac{\lambda_1}{\lambda_3} = \frac{V_3}{V_1}$$

$$\frac{\lambda_1}{\lambda_3} = \frac{Z_H}{n_3} \times \frac{n_1}{Z_H} = \frac{1}{3} \Rightarrow \frac{\lambda_3}{\lambda_1} = \frac{3}{1}$$

2. Ans (5)

$$13.6 Z^2 \left[\frac{1}{4} - \frac{1}{9} \right] = 47.2$$

$$Z^2 \approx 25$$

$$Z \approx 5.00$$

3. Ans (412)

$$eV_s = hf - \phi$$

$$v_s = \frac{h}{e} f - \phi$$

$$\text{Slope} = \frac{h}{e} = 4.12 \times 10^{-15}$$

$$h = 4.12 \times 10^{-15} \text{ eVs}$$

4. Ans (1)

$$T = 2\pi \sqrt{\frac{\mu}{K}}$$

$$\& t_1 = \frac{T}{4}, \frac{3T}{4}, \frac{5T}{4}, \dots$$

5. Ans (582)

$$y_{\text{net}} = y_1 + y_2 + y_3$$

$$= a [\sin \omega t + \sin(\omega t + 45^\circ) + \sin(\omega t + 90^\circ)]$$

$$= (\sqrt{2} + 1) a \sin(\omega t + 45^\circ)$$

Energy ratio is

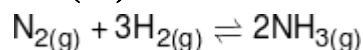
$$\frac{E_{\text{resultant}}}{E_{\text{ring}}} = \left(\frac{A}{a} \right)^2 = (\sqrt{2} + 1)^2 = (3 + 2\sqrt{2})$$

$$E_{\text{resultant}} = 5.82 \text{ J}$$

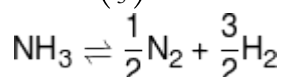
PART 2 : CHEMISTRY

SECTION-I

1. **Ans (D)**



$$K_C = \frac{\left(\frac{4}{3}\right)^2}{\left(\frac{1}{3}\right)(1)} \times (\sqrt{3})^2 = 16$$



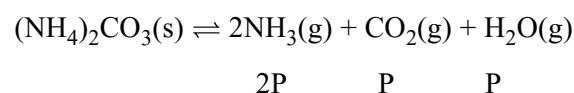
$$K_C^1 = \frac{1}{4}$$

2. **Ans (A)**

$$K_P = K_C (RT)^{\Delta n_g}$$

here $\Delta n_g = 2$

3. **Ans (B)**



$$K_P = (2P)^2 \cdot P \cdot P = 64 \text{ atm}$$

$$P = 2 \text{ atm}$$

$$P_{\text{total}} = 2P + P + P = 8 \text{ atm}$$

4. **Ans (C)**

$$K_a \times K_b = K_w$$

5. **Ans (D)**

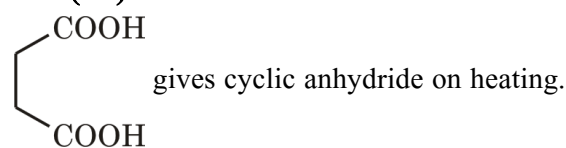
$$\text{Solubility of MX} = (4 \times 10^{-10})^{1/2}$$

$$\begin{aligned} \text{MX}_3 : K_{sp} &= 27s^4 \\ &= 27 (2 \times 10^{-5})^4 \\ &= 4.32 \times 10^{-18} \end{aligned}$$

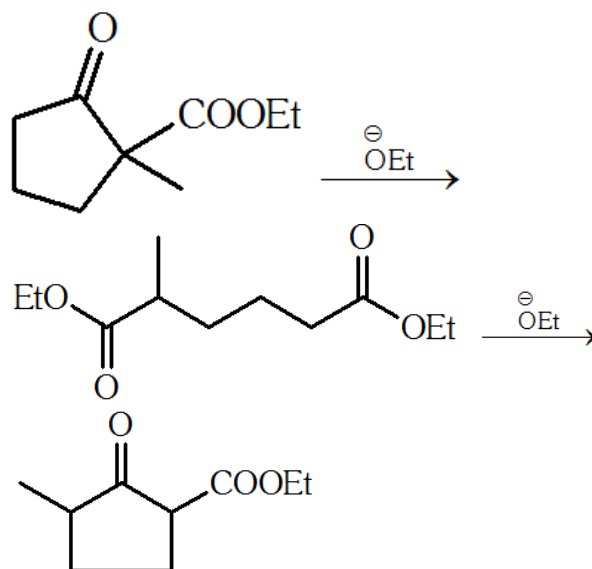
6. **Ans (C)**

Acid halide is most reactive due to -I of Cl.

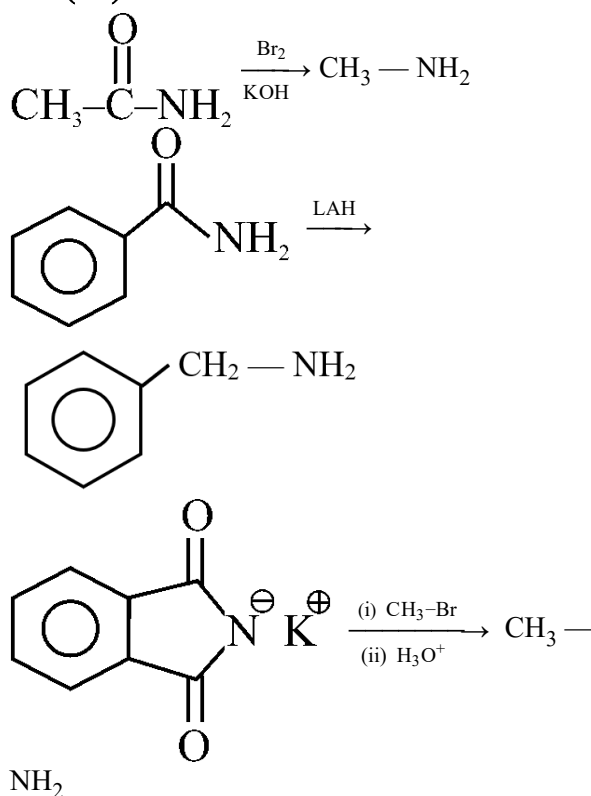
7. **Ans (D)**



8. **Ans (C)**



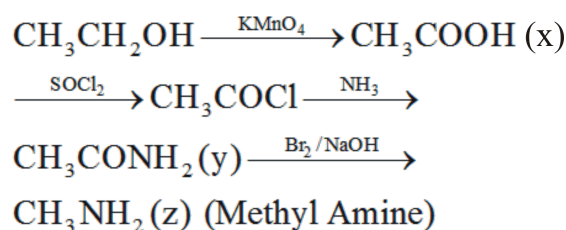
9. **Ans (D)**



NH₂

All primary amines give isosynide test

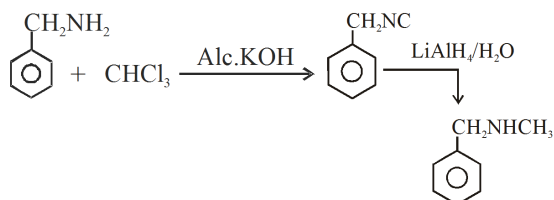
10. **Ans (C)**



11. **Ans (D)**

In azo-coupling reaction diazonium ion acts as an electrophile. The presence of NO_2 group makes the diazo group more electron deficient.

12. **Ans (B)**



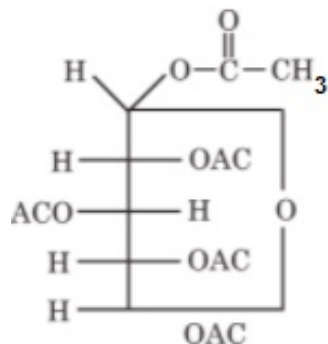
13. **Ans (C)**

Milk sugar (lactose) C1 - C4 β -glycosidic linkage is present.

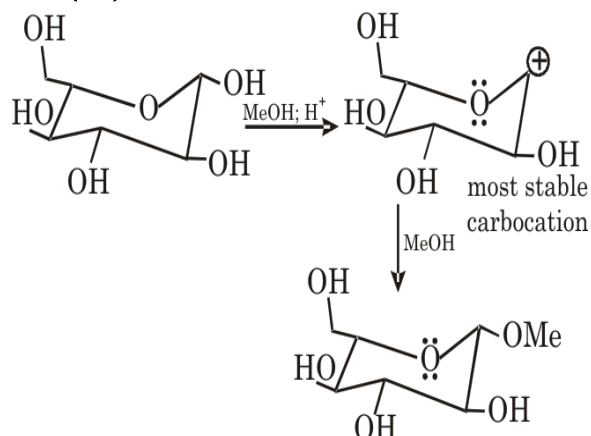
14. **Ans (D)**

There is no hemiacetal linkage in pentacetate of glucose. So, No -CHO group in alkaline medium.

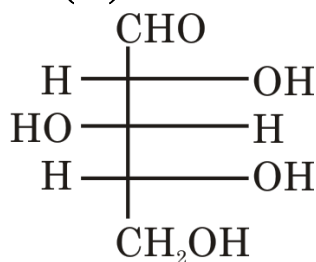
Pentacetate of glucose is



15. **Ans (A)**

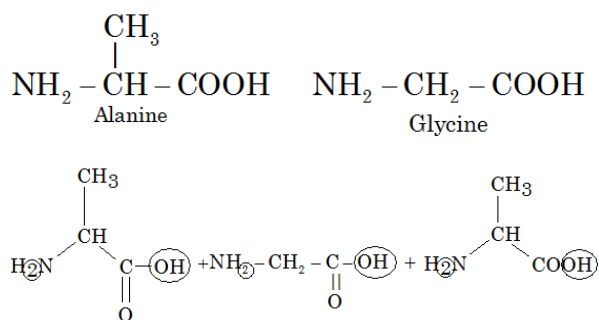


16. **Ans (B)**

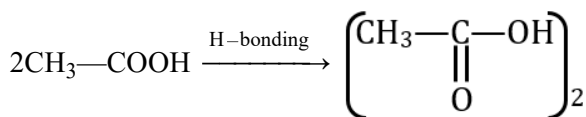


is a meso compound.

17. **Ans (C)**



18. **Ans (B)**

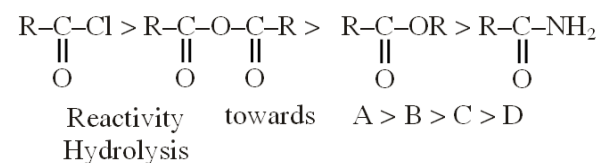


19. **Ans (C)**

k_a $\propto$ acid strength

$$\text{Acid strength} \propto \frac{-M, -H, -I}{+M, +H, +I}$$

20. **Ans (A)**



PART 2 : CHEMISTRY

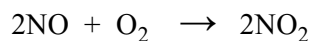
SECTION-II

1. **Ans (1)**

$$\Delta n_g = 1$$

$$\text{So, } \frac{K_P}{K_C} = (RT)^1$$

2. Ans (2)



$$\begin{array}{ccc} 2 & 0.5 & 0 \\ 2-1 & 0 & 1 \end{array}$$

$$n_i = 2.5, n_f = 2$$

$$\Delta n = 0.5$$

$$\Delta P = \frac{\Delta n RT}{v} = \frac{0.5 \times 0.0821 \times 300}{6.25} = 2$$

3. Ans (6)

$$K_h = \frac{x^2}{C} \Rightarrow x = [\text{H}^{\oplus}] = \sqrt{K_h \cdot C}$$

$$[\text{H}^{\oplus}] = \sqrt{10^{-9} \times 0.001} = 10^{-6} \text{ M}$$

$$\text{pH} = 6$$

4. Ans (4)

Functional group present are

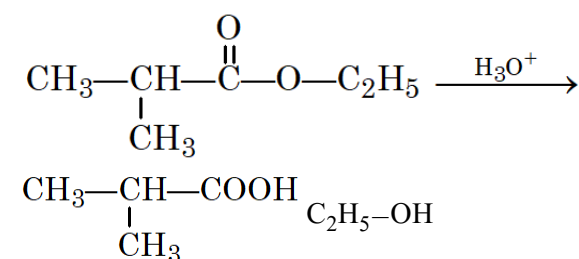
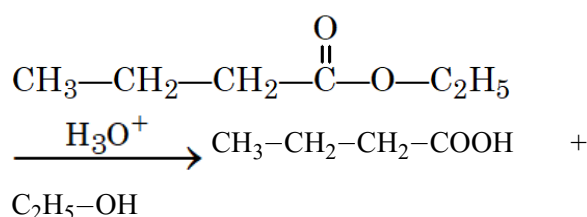
—COCH₃ (ester)

—COOH (Acid)

—NH₂ (Primary Amine)

—CONH₂ (Amide)

5. Ans (2)



PART 3 : MATHEMATICS

SECTION-I

1. Ans (C)

Given lines are parallel

Hence dist. between lines = diameter of circle

$$\text{Hence } r = \frac{1}{2} (\text{dist. between lines})$$

$$= \frac{1}{2} \left| \frac{10 - (-15)}{\sqrt{3^2 + 4^2}} \right| = \frac{5}{2}$$

2. Ans (A)

The position vector of any point at t is

$$\vec{r} = (2 + t^2)\hat{i} + (4t - 5)\hat{j} + (2t^2 - 6t)\hat{k}$$

$$\Rightarrow \frac{d\vec{r}}{dt} = 2t\hat{i} + 4\hat{j} + (4t - 6)\hat{k}$$

$$\Rightarrow \left. \frac{d\vec{r}}{dt} \right|_{t=2} = 4\hat{i} + 4\hat{j} + 2\hat{k}$$

$$\text{and } \left| \left. \frac{d\vec{r}}{dt} \right|_{t=2} \right| = \sqrt{16 + 16 + 4} = 6$$

Hence, the required unit tangent vector at t = 2 is

$$\frac{1}{3} (2\hat{i} + 2\hat{j} + \hat{k})$$

3. Ans (C)

Let $\alpha, \beta, \gamma \rightarrow$ angles made by direction continues with x, y, z - axis $\alpha = \beta = \gamma$

$$\alpha = \beta = \gamma$$

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$3\cos^2 \alpha = 1; \cos \alpha = \pm \frac{1}{\sqrt{3}}$$

$$p = (2, -1, 2)$$

$$\text{Equation of line } \frac{x-2}{\frac{1}{\sqrt{3}}} = \frac{y+1}{\frac{1}{\sqrt{3}}} = \frac{z-2}{\frac{1}{\sqrt{3}}} = k$$

$$Q^x = \left(\frac{k}{\sqrt{3}} + 2, \frac{k}{\sqrt{3}} - 1, \frac{k}{\sqrt{3}} + 2 \right)$$

$$2 \left(\frac{k}{\sqrt{3}} + 2 \right) + \frac{k}{\sqrt{3}} - 1 + \frac{k}{\sqrt{3}} + 2 = 9$$

$$\frac{4k}{\sqrt{3}} + 5 = 9$$

$$\frac{4k}{\sqrt{3}} = 4 \Rightarrow k = \sqrt{3}$$

$$\text{Point } Q \equiv (3, 0, 3)$$

$$P = (2, -1, 2)$$

$$PQ = \sqrt{1 + 1 + 1} = \sqrt{3}$$

4. Ans (B)

Given circles will intersect orthogonally,

$$\text{if } 2(1 \times 0 + k \times k) = 6 + k$$

$$[\text{Using : } 2(g_1 g_2 + f_1 f_2) = c_1 + c_2]$$

$$\Rightarrow 2k^2 - k - 6 = 0$$

$$\Rightarrow (2k + 3)(k - 2) = 0$$

$$\Rightarrow k = 2, -\frac{3}{2}$$

5. Ans (B)

Let $\vec{a} = \lambda\vec{b} + \mu\vec{c}$

$\vec{a}$ is equally inclined to $\vec{b}$ and $\vec{d}$, where

$\vec{d} = \hat{j} + 2\hat{k}$.

$$\Rightarrow \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|} = \frac{\vec{a} \cdot \vec{d}}{|\vec{a}||\vec{d}|}$$

$$\Rightarrow \frac{(\lambda\vec{b} + \mu\vec{c}) \cdot \vec{b}}{(\lambda\vec{b} + \mu\vec{c}) \cdot \vec{d}} = \frac{(\lambda\vec{b} + \mu\vec{c}) \cdot \vec{b}}{(\lambda\vec{b} + \mu\vec{c}) \cdot \vec{d}}$$

$$= \frac{\left[\lambda(2\hat{i} + \hat{j}) + \mu(\hat{i} - \hat{j} + \hat{k}) \right] \cdot (2\hat{i} + \hat{j})}{\left[\lambda(2\hat{i} + \hat{j}) + \mu(\hat{i} - \hat{j} + \hat{k}) \right] \cdot (\hat{j} + 2\hat{k})}$$

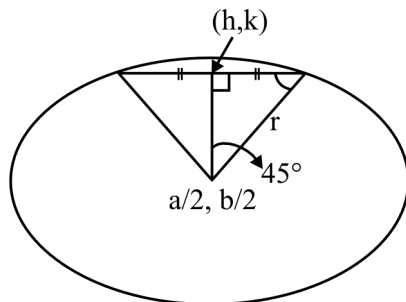
$$= \frac{\sqrt{5}}{\sqrt{5}}$$

or $\lambda(4 + 1) + \mu(2 - 1) = \lambda(1) + \mu(-1 + 2)$

or $4\lambda = 0$, i.e., $\lambda = 0$

$\therefore \hat{a} = \frac{\hat{i} - \hat{j} + \hat{k}}{\sqrt{3}}$

6. Ans (D)



$e = \sqrt{\frac{a^2}{4} + \frac{b^2}{4}} = \frac{\sqrt{a^2 + b^2}}{2}$

$\cos 45^\circ = \frac{\sqrt{\left(4 - \frac{a}{2}\right)^2 + \left(k - \frac{b}{2}\right)^2}}{\frac{\sqrt{a^2 + b^2}}{2}}$

$\left(\frac{1}{\sqrt{2}}\right)^2 = \frac{1}{4} \left(\frac{(2x - a)^2 + (2y - b)^2}{\frac{(a^2 + b^2)}{4}} \right)$

$\frac{a^2 + b^2}{2} = 4x^2 + 4y^2 - 4ax - 4by + a^2 + b^2$

$4x^2 + 4y^2 - 4ax - 4by + \frac{a^2 + b^2}{2} = 0$

$x^2 + y^2 - ax - by + \frac{a^2 + b^2}{8} = 0$

7. Ans (C)

Given that $\vec{a}$, $\vec{b}$ and $\vec{c}$ are non-coplanar. Thus,

$[\vec{a} \vec{b} \vec{c}] \neq 0 \quad \dots(i)$

Again $\vec{a} \times (\vec{b} \times \vec{c}) \cdot (\vec{a} \times \vec{c}) = 0$

or $[(\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}] \cdot (\vec{a} \times \vec{c}) = 0$

or $(\vec{a} \cdot \vec{c})[\vec{b} \vec{a} \vec{c}] = 0$

or $(\vec{a} \cdot \vec{c}) = 0$

Hence, $\vec{a}$ and $\vec{c}$ are perpendicular (ii)

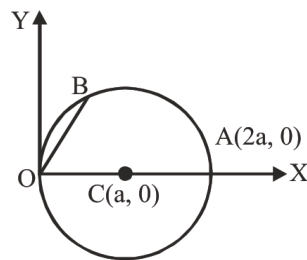
$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$

or $[\vec{a} \times (\vec{b} \times \vec{c})] \times \vec{c} = \vec{0}$

8. Ans (B)

Here the equation of circle is

$(x - a)^2 + (y - 0)^2 = a^2 \Rightarrow x^2 + y^2 - 2ax = 0$



Now the point of intersection of circle and chord i.e.,

Put $y = mx$ in equation of circle and solve it.

O and B are $O(0, 0)$ and $B\left(\frac{2a}{1 + m^2}, \frac{2am}{1 + m^2}\right)$.

Hence the equation of circle (as chord OB as diameter) is $(x^2 + y^2)(1 + m^2) - 2a(x + my) = 0$.

9. Ans (D)

Let $\vec{a} = \vec{i}$, $\vec{b} = \vec{j}$, $\vec{c} = \vec{k}$

$$\begin{cases} \vec{r}_1 = p\vec{a} + q\vec{b} + \vec{c} = p\vec{i} + q\vec{j} + \vec{k} \\ \vec{r}_2 = \vec{a} + p\vec{b} + q\vec{c} = \vec{i} + p\vec{j} + q\vec{k} \end{cases}$$

$$\begin{cases} \vec{r}_1 + 2\vec{r}_2 = \hat{i}(p + 2) + \hat{j}(q + 2p) + \hat{k}(1 + 2q) \\ 2\vec{r}_1 + \vec{r}_2 = (2p + 1)\hat{i} + (2q + p)\hat{j} + (2 + q)\hat{k} \end{cases}$$

$$(\vec{r}_1 + 2\vec{r}_2) \parallel (2\vec{r}_1 + \vec{r}_2)$$

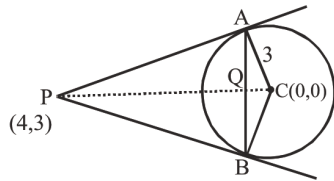
$$\Rightarrow \frac{p + 2}{2p + 1} = \frac{q + 2p}{2q + p} = \frac{1 + 2q}{2 + q}$$

$p = q$ & $q = 1$

10. Ans (C)

Equation of AB is $4x + 3y = 9$ (i)

{ $\because$ it is chord of contact }



$OQ = \frac{9}{5}$ [perpendicular distance of AB from origin]

$$AQ = \sqrt{OA^2 - OQ^2} = \sqrt{9 - \frac{81}{25}} = \frac{12}{5}$$

$$AB = 2AQ = \frac{24}{5}$$

$$PQ = \frac{16 + 9 - 9}{\sqrt{16 + 9}} = \frac{16}{5}$$

$$\text{Hence the area} = \frac{1}{2} \times \frac{24}{5} \times \frac{16}{5} = \frac{192}{25}$$

$$\text{Aliter : Required area} = \frac{a}{h^2 + k^2} (h^2 + k^2 - a^2)^{3/2}$$

$$= \frac{3}{4^2 + 3^2} (42 + 32 - 9)^{3/2} = \frac{192}{25}$$

11. Ans (C)

Suppose the bisector of angle A meets BC at D.

Then AD divides BC in the ratio AB : AC.

So, P.V. of D is given by

$$\frac{|\overrightarrow{AB}|(2\hat{i} + 5\hat{j} + 7\hat{k}) + |\overrightarrow{AC}|(2\hat{i} + 3\hat{j} + 4\hat{k})}{|\overrightarrow{AB}| + |\overrightarrow{AC}|}$$

$$\text{But } \overrightarrow{AB} = -2\hat{i} - 4\hat{j} - 4\hat{k}$$

$$\text{and } \overrightarrow{AC} = -2\hat{i} - 2\hat{j} - \hat{k}$$

$$\Rightarrow |\overrightarrow{AB}| = 6 \text{ and } |\overrightarrow{AC}| = 3$$

Therefore, P.V. of D is given by

$$\frac{6(2\hat{i} + 5\hat{j} + 7\hat{k}) + 3(2\hat{i} + 3\hat{j} + 4\hat{k})}{6 + 3}$$

$$= \frac{1}{3}(6\hat{i} + 13\hat{j} + 18\hat{k})$$

12. Ans (C)

radius ≤ 5

$$\sqrt{\left(\frac{\lambda}{2}\right)^2 + \left(\frac{1-\lambda}{2}\right)^2} - 5 \leq 5$$

$$\Rightarrow 2\lambda^2 - 2\lambda - 119 \leq 0$$

$$\Rightarrow \frac{1 - \sqrt{239}}{2} \leq \lambda \leq \frac{1 + \sqrt{239}}{2}$$

$$\Rightarrow -7.2 \leq \lambda \leq 8.2 \text{ (nearly)}$$

$$\Rightarrow \lambda = -7, -6, \dots, 8$$

13. Ans (D)

Here, the required plane is

$$a(x - 4) + b(y - 3) + c(z - 2) = 0$$

$$\text{Also } a + b + 2c = 0 \text{ and } a - 4b + 5c = 0$$

Solving, we have

$$\frac{a}{5 + 8} = \frac{b}{2 - 5} = \frac{c}{-4 - 1} = k$$

$$\frac{a}{13} = \frac{b}{-3} = \frac{c}{-5} = k$$

Therefore, the required equation of plane is

$$-13x + 3y + 5z + 33 = 0$$

14. Ans (C)

$$\vec{r} = (1 + \lambda - \mu)\hat{i} + (2 - \lambda)\hat{j} + (3 - 2\lambda + 2\mu)\hat{k}$$

$\Rightarrow$

$$\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(\hat{i} - \hat{j} - 2\hat{k}) + \mu(-\hat{i} + 2\hat{k})$$

which is a plane passing through

$$\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k} \text{ and parallel to the vectors}$$

$$\vec{b} = \hat{i} - \hat{j} - 2\hat{k} \text{ and } \vec{c} = -\hat{i} + 2\hat{k} \text{ therefore, it}$$

$$\text{perpendicular to the vector}$$

$$\vec{n} = \vec{b} \times \vec{c} = -2\hat{i} - \hat{k}$$

Hence, its vector equation is

$$(\vec{r} - \vec{a}) \cdot \vec{n} = 0$$

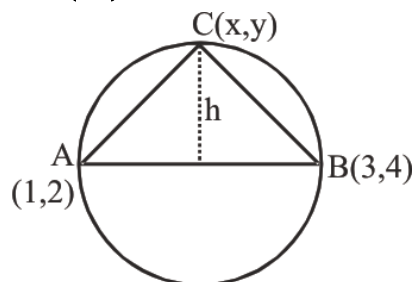
$$\Rightarrow \vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n} \Rightarrow \vec{r} \cdot (-2\hat{i} - \hat{k}) = -2 - 3$$

$$\Rightarrow \vec{r} \cdot (2\hat{i} + \hat{k}) = 5$$

so, the cartesian equation is

$$(x\hat{i} + y\hat{j} + z\hat{k}) \cdot (2\hat{i} + \hat{k}) = 5 \text{ or } 2x + z = 5$$

15. Ans (B)



$\because$ A & B are end's of diameter, diameter $= 2\sqrt{2}$

$\therefore$ radius $= \sqrt{2}$

Let height of ΔABC is h.

$$\text{Now, } \frac{1}{2} \cdot \text{Base} \times h = 1 \quad \{\text{Base} = \text{diameter}\}$$

$$\Rightarrow \frac{1}{2} \times 2\sqrt{2} \times h = 1$$

$$\Rightarrow h = \frac{1}{\sqrt{2}}$$

$\because h < r$

therefore no. of position of C is 4.

16. Ans (B)

Let the components of the line segment vector be a, b, c then

$$a^2 + b^2 + c^2 = (63)^2 \quad \dots(i)$$

$$\text{also. } \frac{a}{3} = \frac{b}{-2} = \frac{c}{6} = \lambda \text{ (say)}$$

$$\Rightarrow a = 3\lambda, b = -2\lambda, c = 6\lambda$$

$$\text{from (i)} \quad 9\lambda^2 + 4\lambda^2 + 36\lambda^2 = (63)^2$$

$$\Rightarrow 49\lambda^2 = (63)^2$$

$$\Rightarrow \lambda = \pm \frac{63}{7} = \pm 9$$

$$\because a = 3\lambda < 0$$

$\therefore$ line makes obtuse angle with x-axis

$$\Rightarrow \lambda = -9$$

So. reqd. components are $-27, 18, -54$

17. Ans (A)

Let the equation be $x^2 + y^2 + 2gx + 2fy + c = 0$ it passes through $(-1, -3)$ and $(3, 0)$ therefore

$$10 - 2g - 6f + c = 0 \quad \dots(i)$$

$$9 + 6g + c = 0 \quad \dots(ii)$$

Slope of tangent $= -4/3$

$$\left(\frac{0+f}{3+g} \right) \left(\frac{-4}{3} \right) = -1$$

$$\Rightarrow 3g - 4f + 9 = 0 \quad \dots(iii)$$

$$\text{solving } g = -1, f = 3/2, c = -3$$

18. Ans (B)

Let the equation of line passes through

$$\frac{x-1}{a} = \frac{y-0}{b} = \frac{z+1}{c} \quad \dots(1)$$

It is perpendicular to two given lines

$$\therefore 2a + 7b - 3c = 0 \quad \dots(2)$$

$$\text{and } 2a - 2b + 5c = 0 \quad \dots(3)$$

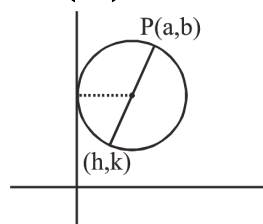
from (2) & (3)

$$\frac{a}{29} = \frac{b}{-16} = \frac{c}{-18}$$

$\therefore$ Equation of line

$$\frac{x-1}{29} = \frac{y-0}{-16} = \frac{z+1}{-18}$$

19. Ans (D)



$$\text{centre } \left(\frac{h+a}{2}, \frac{k+b}{2} \right)$$

$$\text{radius} = \frac{1}{2} \sqrt{(h-a)^2 + (k-b)^2}$$

Acc. to condition

$$\frac{1}{2} \sqrt{(h-a)^2 + (k-b)^2} = \frac{h+a}{2}$$

$$\Rightarrow (k-b)^2 = 4ah$$

$$\Rightarrow (y-b)^2 = 4ax$$

20. Ans (A)

$$\text{Let plane is } \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1 \quad \dots(i)$$

mid point of $P(1, 2, 3)$ and $Q(-3, 4, 5)$

i.e., $-1, 3, 4$ lie on Eq. (i)

$$\therefore -\frac{1}{a} + \frac{3}{b} + \frac{4}{c} = 1 \quad \dots(ii)$$

Also, PQ is parallel to normal of the plane (i)

$$\frac{1/a}{-4} = \frac{1/b}{2} = \frac{1/c}{2}$$

$$\Rightarrow \frac{1}{-2a} = \frac{1}{b} = \frac{1}{c} = \lambda \text{ (say)}$$

$$\therefore \frac{1}{a} = -2\lambda, \frac{1}{b} = \lambda, \frac{1}{c} = \lambda$$

$$\therefore \text{From Eq. (ii), } 2\lambda + 3\lambda + 4\lambda = 1$$

$$\therefore \lambda = \frac{1}{9}$$

$$a = -\frac{1}{2\lambda}, b = \frac{1}{\lambda}, c = \frac{1}{\lambda}$$

$$a = -\frac{9}{2}, b = 9, c = 9$$

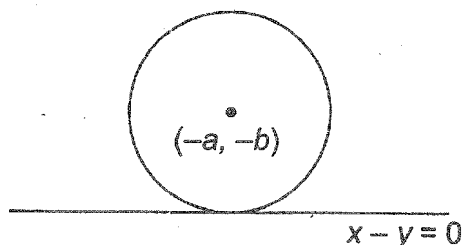
$$\text{Intercepts are } \left(-\frac{9}{2}, 9, 9 \right).$$

PART 3 : MATHEMATICS

SECTION-II

1. Ans (1)

Radical axis, $x - y = 0$



Centre of 1st circle is $(-a, -b)$ and $r = \sqrt{a^2 + b^2 - c}$.

Now perpendicular distance from center on radical axis = radius of the circle

$$(a - b)^2 = 2[a^2 + b^2 - c]$$

$$\Rightarrow a^2 + b^2 - 2ab = 2[a^2 + b^2 - c]$$

$$a^2 + b^2 + 2ab = 2c$$

$$\Rightarrow \frac{(a + b)^2}{2c} = 1$$

2. Ans (2)

$$\text{Given, } \frac{(\vec{b} \cdot \vec{a}) \vec{a}}{|\vec{a}|^2} = \frac{4}{3} (\hat{i} - \hat{j} - \hat{k})$$

$$\Rightarrow \frac{\{(\lambda \hat{i} - 3\hat{j} + \hat{k}) \cdot (\hat{i} - \hat{j} - \hat{k})\} (\hat{i} - \hat{j} - \hat{k})}{(1 + 1 + 1)}$$

$$= \frac{4}{3} (\hat{i} - \hat{j} - \hat{k})$$

$$\Rightarrow (\lambda + 3 - 1) (\hat{i} - \hat{j} - \hat{k}) = 4 (\hat{i} - \hat{j} - \hat{k})$$

$$\Rightarrow (\lambda + 2) (\hat{i} - \hat{j} - \hat{k}) = 4 (\hat{i} - \hat{j} - \hat{k})$$

On equating the coefficient of $\hat{i}$, we get

$$\lambda + 2 = 4 \Rightarrow \lambda = 2$$

3. Ans (20)

Since, the given line touches the given circle, the length of the perpendicular from the centre $(2, 4)$ of the circle to the line $3x - 4y - k = 0$ is equal to the radius $\sqrt{4 + 16 + 5} = 5$ of the circle.

$$\therefore \frac{3 \times 2 - 4 \times 4 - k}{\sqrt{9 + 16}} = \pm 5$$

$$\Rightarrow k = 15 \quad [\because k > 0]$$

hence equation of tangent is

$$3x - 4y - 15 = 0 \quad \dots (1)$$

Let equation of normal to circle $4x + 3y = \lambda$

It passes through centre $(2, 4) \Rightarrow \lambda = 20$

hence equation of normal is

$$4x + 3y = 20 \quad \dots (2)$$

Solve (1) & (2)

$$a = 5, b = 0; k + a + b = 15 + 5 + 0 = 20$$

4. Ans (3)

$$\text{Shortest distance} = \frac{|\vec{a} - \vec{c} \cdot \vec{b} \times \vec{d}|}{|\vec{b} \times \vec{d}|}$$

$$\text{Here } \vec{a} = 3\hat{i} + 8\hat{j} + 3\hat{k}, \vec{b} = 3\hat{i} - \hat{j} + \hat{k}, \vec{c} = -3\hat{i} - 7\hat{j} + 6\hat{k}, \vec{d} = -3\hat{i} + 2\hat{j} + 4\hat{k}$$

$$\therefore \text{Shortest distance} = \frac{270}{\sqrt{270}} = \sqrt{270} = 3\sqrt{30} = \lambda\sqrt{30} \Rightarrow \lambda = 3$$

5. Ans (6)

Let the equation of the circle be

$$(x - a)^2 + (y - a)^2 = a^2, a > 0,$$

It touches $4x + 3y - 12 = 0$

$$\therefore \left| \frac{4a + 3a - 12}{5} \right| = a$$

$$7a - 12 = \pm 5a$$

take '+' sign take '-' sign

$$2a = 12 \quad 12a = 12$$

$$a = 6 \quad a = 1$$

$\therefore$ radius of larger circle $a = 6$